

**Predictive Valuation Modeling for Detecting Price Anomalies in Residential
Real Estate Listings
BUSINESS ANALYTICS
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SADHANA T P
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1) Problem Statement and Objectives

1.1) Problem Statement

Residential property listings can vary substantially in price even when properties have similar characteristics. Factors such as built-up area, number of bedrooms, bathrooms, parking availability, location, furnishing, floor, property age, and construction status can influence the expected listing price. For a buyer or property platform, it can therefore be difficult to determine whether an asking price is broadly consistent with the characteristics of a property.

This case study addresses the following business problem,

Can the expected listing price of a Bengaluru residential resale apartment be predicted from its property characteristics, and can substantial differences between the actual listing price and the predicted price be used to identify potentially underpriced and potentially overpriced listings?

The objective is not to determine the actual market value of a property. Instead, the analysis provides a data-driven screening mechanism that identifies listings whose observed asking prices differ substantially from the prices expected by the predictive model. Such listings can then be examined further using property-specific information that may not be captured in the dataset.

1.2) Objectives

The study has the following specific objectives:

- To collect and prepare a large dataset of Bengaluru residential resale property listings from publicly accessible online listings and perform appropriate data cleaning and preprocessing for analysis.
- To develop and evaluate predictive regression models for estimating residential property listing prices using relevant property characteristics such as area, BHK, bathrooms, parking, location, furnishing, floor, age, and construction status.
- To identify potentially underpriced and potentially overpriced listings by comparing actual listing prices with model-predicted prices and applying a statistical threshold to the resulting price deviations.

These objectives support the broader business purpose of providing a property-price screening approach that can help buyers, sellers, and real-estate platforms identify listings that may require additional investigation.

2) Data Collection and Dataset Description

2.1) Data Collection

The dataset for this case study was collected from Housing.com, a publicly accessible online real-estate listing platform^[1]. The study focuses on residential resale properties in Bengaluru. Web scraping was used as the primary data collection method to obtain a large number of real-world property listings containing asking prices and relevant property characteristics.

The data was collected using the Housing.com Property Scraper available through Apify^[2](<https://apify.com/shahidirfan/housing-com-property-scraper>). The scraping process was configured to collect residential properties listed for sale in Bengaluru, with the focus on resale listings. Multiple scraping runs were performed to obtain sufficient records for the case study. The final collection resulted in 10,000 property listings.

The collected data was then consolidated and checked for data quality. Duplicate listing identifiers and duplicate URLs were examined, and the records were restricted to the intended Bengaluru residential property scope. Relevant attributes were extracted from the structured listing information, property features, titles, and descriptions where necessary.

2.2) Dataset Description

The final dataset contains 10,000 Bengaluru residential resale property listings. Each record represents an individual property listing and contains information describing its price, physical characteristics, location, and other property attributes.

The major variables in the dataset are,

Variable	Description
listingId	Unique identifier of the property listing
title	Title of the property listing
price	Asking price of the property and prediction target
area_sqft	Property area in square feet
bhk	Number of bedrooms
bathrooms	Number of bathrooms
parking	Number of parking spaces
location	Locality of the property
latitude	Geographic latitude
longitude	Geographic longitude
floor_number	Floor on which the property is located
total_floors	Total number of floors in the building
furnishing	Furnishing status
facing	Direction the property faces
age_years	Age of the property in years
possession_category	Possession status/category
possession_date	Possession date, where available

is_under_construction	Indicates whether the property is under construction
project_name	Name of the residential project, where available
posted_date	Listing posting date
propertyType	Property type
saleTag	Sale classification

2.3) Dataset Characteristics

The dataset represents a broad range of residential resale properties across Bengaluru. The collected listings include variations in property size, BHK configuration, number of bathrooms, parking availability, locality, furnishing status, floor, property age, and construction status.

The asking price ranges from approximately ₹20 lakh to ₹70 crore, while property area ranges from approximately 300 to 9,300 sq.ft. The median property area is approximately 1,350 sq.ft., and the median asking price is approximately ₹1.58 crore.

The dataset contains properties with different bedroom configurations, with 2-BHK and 3-BHK properties forming the largest groups. It also covers a large number of Bengaluru localities, providing geographical variation that can be incorporated into the price prediction model.

The resulting dataset provides sufficient variation in property characteristics and asking prices for performing exploratory analysis, predictive modeling, and price anomaly detection.

2.4) Data Collection Summary

Aspect	Description
Data source	housing.com ^[1]
Collection method	Web scraping
Scraping platform	Apify ^[2]
Dataset domain	Residential real estate
Geographic scope	Bengaluru
Property category	Residential resale
Final records	10,000
Target variable	Property asking price
Primary use	Price prediction and price anomaly detection

3) Data Preparation and Exploratory Analysis

3.1) Data Preparation

The collected property listings were cleaned and transformed before analysis to obtain a consistent analytical dataset of 10,000 Bengaluru residential resale apartment listings. The major cleaning

steps included cleaning property prices and area values, extracting and resolving BHK, bathroom, and parking information, converting coordinates into latitude and longitude, parsing floor information into floor number and total floors, converting property age into numeric years, standardizing furnishing/facing and possession information, and performing duplicate and validity checks.

Missing values were handled during the machine-learning preprocessing stage. Median imputation was used for numerical variables and most-frequent imputation for categorical variables. Categorical variables were then transformed using one-hot encoding^[6]. The preprocessing pipeline was fitted using the training data to reduce data leakage.

For modeling, the selected numerical variables were area_sqft, bhk, bathrooms, parking, floor_number, total_floors, and age_years. The categorical variables were location, furnishing, facing, possession_category, and is_under_construction. The derived price_per_sqft variable was used only for descriptive analysis and was excluded from modeling because it is directly calculated from the target price.

3.2) Exploratory Data Analysis

EDA was performed to understand the distribution of property characteristics and their relationships with listing price.

3.2.1) Listing Price Distribution



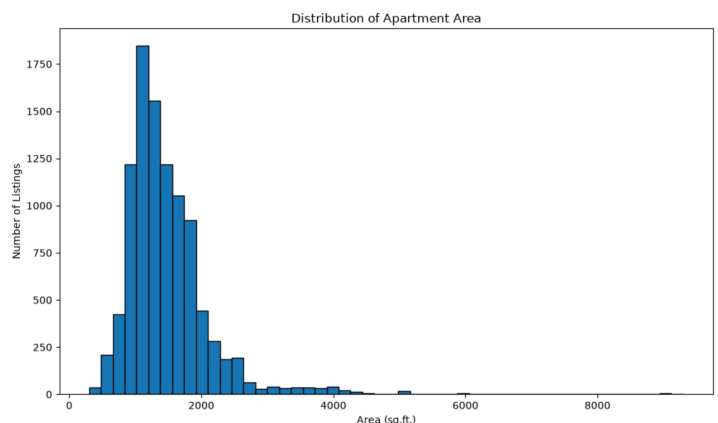
Figure 1) Distribution of Bengaluru Resale Apartment Prices

The price distribution is strongly right-skewed. Most listings are concentrated in the lower price range, while a smaller number of high-value properties extend the upper tail. The median asking price is approximately ₹1.587 crore, compared with a mean of approximately ₹1.938 crore. Prices range from approximately ₹20 lakh to ₹70 crore.

3.2.2) Apartment Area Distribution

Figure 2) Distribution of Apartment Area

Most properties have areas concentrated between approximately 1,000 and 2,000 sq.ft. The median area is approximately 1,342 sq.ft., while the mean is approximately 1,484 sq.ft. A smaller number of large properties creates a right-skewed distribution.



3.2.3) BHK Distribution

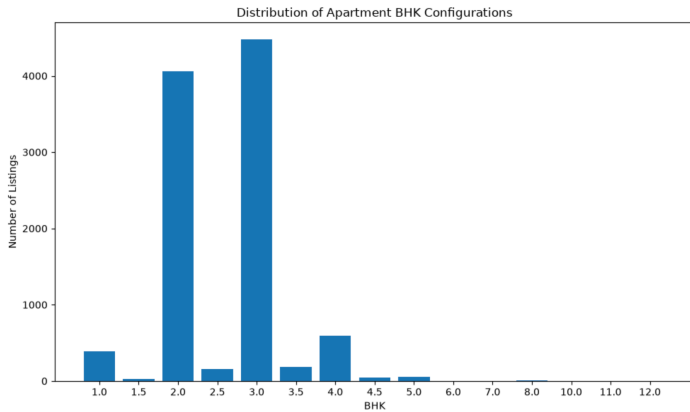


Figure 3) Distribution of Apartment BHK Configurations

The dataset is dominated by 2-BHK and 3-BHK apartments. Approximately 4,065 listings are 2-BHK and 4,475 are 3-BHK. Other configurations occur less frequently, with very high BHK categories represented by only a small number of properties.

3.2.4) Area and Listing Price

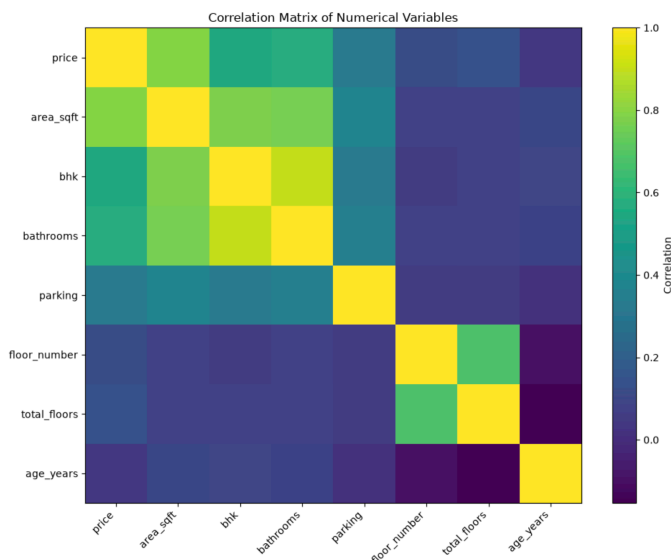
Figure 4) Apartment Area vs Listing Price

A clear positive relationship can be observed between apartment area and listing price. Larger apartments generally have higher asking prices, although considerable variation exists among properties with similar areas. The correlation between area and price is approximately 0.80, making area the strongest numerical variable associated with price.



3.2.5) Correlation Analysis

Figure 5) Correlation Matrix of Numerical Variables

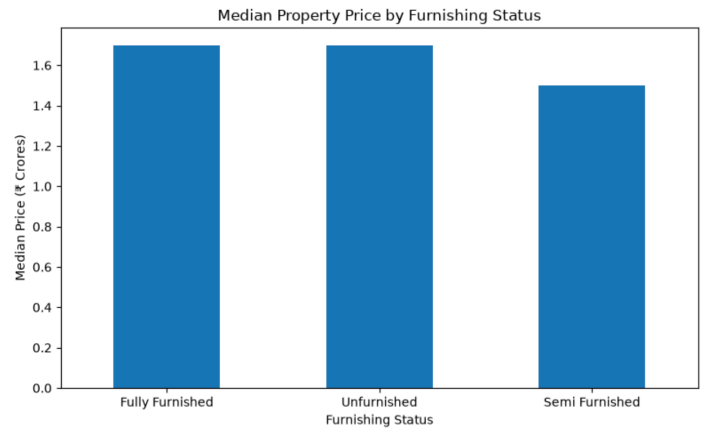


Area has the strongest correlation with price (0.80), followed by bathrooms (0.56) and BHK (0.55). Parking has a weaker positive relationship (0.32), while floor number, total floors, and property age have relatively weak individual correlations with price. The correlation analysis also shows a strong relationship between BHK and bathrooms, which is expected because larger apartment configurations generally contain more bathrooms.

3.2.6) Furnishing Status and Price

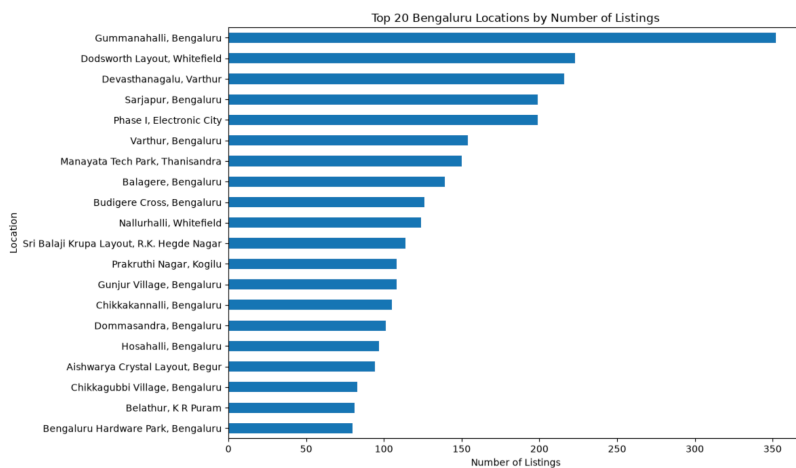
Figure 6) Median Property Price by Furnishing Status

The median prices of fully furnished and unfurnished properties are relatively similar, while semi-furnished properties have a somewhat lower median in this dataset. This suggests that furnishing status alone does not provide a strong explanation for price differences and should be considered together with other property characteristics.



3.2.7) Location Distribution

Figure 7) Top 20 Bengaluru Locations by Number of Listings



The dataset covers a wide range of Bengaluru locations. Gummanahalli, Dodsworth Layout–Whitefield, Devasthanagalu–Varthur are among the locations with the largest number of listings. The variation in listing counts demonstrates the geographical coverage of the dataset and supports the inclusion of location as a categorical predictive variable.

4) Analytics Method and Implementation

4.1) Analytical Method

The study uses predictive regression analysis to estimate the expected asking price of residential properties. Regression is suitable because the target variable, property listing price, is continuous, while the dataset contains both numerical and categorical property characteristics.

Three regression methods were evaluated, Linear Regression, Random Forest Regression and Gradient Boosting Regression

Linear Regression was used as the baseline model, while Random Forest and Gradient Boosting were included to compare the performance of a linear approach with tree-based ensemble methods.

4.2) Model Implementation

The dataset was divided into 80% training data and 20% testing data using a fixed random state. Numerical missing values were handled using median imputation, while categorical missing values were replaced using the most frequent category. Categorical variables were then converted using one-hot encoding through a scikit-learn preprocessing pipeline^[6].

The predictive features consisted of,

- Numerical: area, BHK, bathrooms, parking, floor number, total floors, and property age.
- Categorical: location, furnishing, facing, possession category, and construction status.

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score^[6]. The results were,

Model	MAE	RMSE	R^2
Linear Regression	₹4.089M	₹6.695M	0.7970
Random Forest	₹3.993M	₹8.891M	0.6419
Gradient Boosting	₹4.256M	₹9.200M	0.6166

Linear Regression was selected because it achieved the highest R^2 and lowest RMSE among the evaluated models. Although Random Forest had a slightly lower MAE, its RMSE and R^2 were substantially weaker.

4.3) Price Deviation and Anomaly Detection

After selecting Linear Regression, 5-fold out-of-fold (OOF) predictions were generated for all 10,000 properties^[6]. Each property's prediction was produced by a model that had not been trained on that property, providing a more appropriate basis for measuring price deviations across the complete dataset.

The price deviation was calculated as:

$$\text{Deviation (\%)} = [(\text{Actual Price} - \text{Predicted Price}) / \text{Actual Price}] \times 100$$

A negative deviation indicates that the observed listing price is below the model prediction, while a positive deviation indicates that it is above the model prediction.

The Interquartile Range (IQR) method was then applied to the valid deviation values to identify unusually large deviations. The resulting thresholds were:

- $Q1 = -16.83\%$
- $Q3 = 13.65\%$
- $IQR = 30.48\%$
- Potentially Underpriced: below -62.56%
- Fairly Priced: -62.56% to $+59.37\%$
- Potentially Overpriced: above $+59.37\%$

This approach provides a systematic way to identify listings and their price category.

5) Comparison with State-of-the-Art Methods

Recent research on residential property-price prediction increasingly uses machine-learning and hybrid approaches to capture nonlinear relationships between property characteristics and prices. The following studies were selected because they address the same broad problem of residential/house price prediction using recent machine-learning or advanced regression methods.^{[3][4][5]}

Parameter	Lahmiri, Bekiros & Avdoulas (2023) ^[3]	Sharma, Harsora & Ogunleye (2024) ^[4]	Rodriguez-Serrano (2024/2025) ^[5]
Dataset	Housing-price data from Taiwan	Ames Housing dataset, Iowa, USA; 2,930 records and 82 variables	Public Seattle real-estate price dataset

Method Used	Boosting ensemble regression trees, Support Vector Regression and Gaussian Process Regression, with Bayesian optimization and 10-fold cross-validation	Linear Regression, Random Forest, Support Vector Regression, Multilayer Perceptron and XGBoost, with hyperparameter tuning	Prototype-based learning for nonlinear real-estate valuation and explainability
Evaluation Metric	Four predictive performance metrics	R^2 , adjusted R^2 , MSE, RMSE, MAE and cross-validation score	RMSE, including repeated experimental runs
Key Result	Boosting ensemble regression trees performed best among the methods evaluated in the study.	XGBoost achieved the strongest reported predictive performance in their experimental setting.	The prototype-based approach produced lower prediction error than several comparison models in the reported experiments and provided property-level explainability.
Comparison with Your Work	Uses more advanced nonlinear models and Bayesian optimization. In comparison, this case study uses Linear Regression, Random Forest and Gradient Boosting with a simpler preprocessing and evaluation pipeline. This project additionally uses OOF predictions and IQR-based price-deviation analysis for anomaly screening.	Both studies compare multiple regression techniques. This case study uses a 10,000-record Bengaluru resale-apartment dataset collected through web scraping, while the study uses the Ames dataset. This project selects Linear Regression based on its own test results rather than assuming a complex model will perform better.	Focuses on nonlinear prediction, explainability and direct comparison between properties. This case study instead uses conventional regression models and focuses on identifying substantial deviations between observed and predicted listing prices using OOF prediction and IQR-based anomaly detection.

5.1) Comparison and Relevance

The reviewed studies demonstrate three important directions in current real-estate analytics^{[7][8][9]}. First, ensemble and nonlinear machine-learning methods are widely used to capture relationships that traditional linear models may not represent. Lahmiri et al. use Bayesian optimization with boosting, SVR and Gaussian-process models^[3], while Sharma et al. compare several algorithms including XGBoost^[4]. Second, recent research also places greater emphasis on interpretability and property comparison. Rodriguez-Serrano's prototype-based approach explicitly represents properties using learned reference prototypes and incorporates the idea of direct comparison into the prediction process^[5].

The present case study takes a somewhat different practical approach. Rather than focusing only on maximizing predictive accuracy, it uses a comparatively simple and interpretable Linear Regression model after comparing it with Random Forest and Gradient Boosting. The selected model is then used to generate 5-fold out-of-fold predictions, allowing every listing to receive a prediction from a model that did not train on that listing. The resulting price deviations are analyzed using an IQR-based anomaly detection method.

Thus, the main distinction of this case study is that price prediction is used as an intermediate step for price-anomaly screening, rather than being the final analytical objective. This connects the predictive model directly to the business problem of identifying listings that warrant further investigation.

6) Results, Business Insights and Recommendations

6.1) Model Results

The three regression models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . Linear Regression achieved the highest R^2 and the lowest RMSE among the evaluated models, and was therefore selected as the final predictive model.

Model	MAE	RMSE	R^2
Linear Regression	₹4.089M	₹6.695M	0.7970
Random Forest	₹3.993M	₹8.891M	0.6419
Gradient Boosting	₹4.256M	₹9.200M	0.6166

The selected Linear Regression model was then used to generate 5-fold out-of-fold (OOF) predictions for all 10,000 listings. The difference between actual and predicted prices was used to identify listings with substantial deviations. An IQR-based rule classified the listings into potentially underpriced, fairly priced, and potentially overpriced categories.

6.2) Price Anomaly Results

Price Category	Number of Listings	Percentage
Fairly Priced	9,313	93.13%
Potentially Underpriced	581	5.81%
Potentially Overpriced	98	0.98%
Invalid Prediction	8	0.08%
Total	10,000	100%

The model therefore identified 581 listings as potentially underpriced and 98 listings as potentially overpriced. These categories are screening indicators based on differences between observed and predicted prices, rather than confirmed market valuations.

6.3) Business Insights

1) Potentially underpriced listings — The 581 potentially underpriced listings had a median actual price of ₹95 lakh, compared with a median predicted price of approximately ₹1.85 crore. Their median area was 1,345 sq.ft. and median BHK was 2.5.

2) Potentially overpriced listings — The 98 potentially overpriced listings had a median actual price of ₹1.37 crore, compared with a median predicted price of approximately ₹38.3 lakh. Their median area was 972.5 sq.ft. and median BHK was 2.

3) Location patterns — The location analysis showed that potentially underpriced listings were concentrated in locations such as Phase I, Electronic City (22 listings), Dodsworth Layout, Whitefield (15), Varthur (14), and Sarjapur (12). Some smaller locations had higher anomaly rates, but these should be interpreted cautiously because their total number of listings was small. For example, Kaverappa Layout had 8 potentially underpriced listings out of only 15 listings. For potentially overpriced listings, locations such as Sree Narayana Nagar, Kadagrahara (4), Shriram

Sameeksha, Singapura (3), Terracon Doctors Enclave, Chandapura (3), and Hennagara, Bommasandra (3) showed multiple flagged listings.

4) Furnishing status — The furnishing distribution does not show a clear standalone distinction between the price categories. Semi-furnished properties form the largest group across all three categories, Fairly priced: 52.74%, Potentially overpriced: 67.35%, Potentially underpriced: 61.96%. Therefore, furnishing status alone does not appear to provide a strong explanation for the identified price anomalies.

5) Construction status — The construction-status distribution is also relatively similar across the categories. Approximately 83%–88% of listings in each category were not under construction. Therefore, construction status does not appear to be a strong standalone differentiator of the identified anomaly categories.

6) Property age — Potentially underpriced properties had a median age of 7 years, compared with 5 years for fairly priced properties and 4.1 years for potentially overpriced properties. This indicates that the potentially underpriced group contains relatively older properties. However, this is an observed association and should not be interpreted as evidence that property age causes a listing to be underpriced.

7) BHK distribution — The potentially overpriced group had a higher concentration of 2-BHK properties (54.08%), while the fairly priced and potentially underpriced groups contained larger proportions of both 2-BHK and 3-BHK properties. Since 2-BHK and 3-BHK properties dominate the overall dataset, this finding should be treated as a descriptive pattern rather than a causal explanation of price anomalies.

6.4) Business Recommendations

Based on the results, the following recommendations can be made:

- **Use the model as a screening tool:** Real-estate platforms can use predicted prices to flag listings whose asking prices substantially differ from model expectations.
- **Investigate potentially underpriced listings:** Such listings can be reviewed for factors not captured by the model, including property condition, documentation, seller urgency, maintenance requirements, building quality, amenities and exact micro-location.
- **Review potentially overpriced listings:** Listings with large positive price deviations can be examined for premium characteristics that may not be adequately represented in the available dataset.
- **Monitor location-level patterns:** Repeated anomalies in specific locations can be investigated further using additional local market information and larger samples.
- **Update the model periodically:** Housing prices and market conditions change over time, so the model should be retrained using newer listing data to maintain its usefulness.
- **Use anomaly labels cautiously:** The identified categories should be treated as **potential pricing signals**, not definitive evidence that a property is genuinely underpriced or overpriced.

7) Conclusion and References

7.1) Conclusion

This case study developed a predictive analytics framework for estimating the expected listing price of Bengaluru residential resale apartments and identifying listings with substantial differences between observed and model-predicted prices. The final analytical dataset contained 10,000 residential resale apartment listings, collected from Housing.com and prepared for exploratory and predictive analysis.

Three regression models—Linear Regression, Random Forest, and Gradient Boosting—were evaluated using MAE, RMSE, and R^2 . Linear Regression was selected as the final model based on the evaluation results. Five-fold out-of-fold predictions were then generated for all listings, allowing price deviations to be calculated without using a property's own target value during its prediction.

Using an IQR-based anomaly-detection approach, the analysis identified 581 potentially underpriced listings, 9,313 fairly priced listings, 98 potentially overpriced listings, and 8 invalid predictions. These classifications are intended as screening indicators rather than definitive statements about the actual market value of a property.

The analysis also identified descriptive patterns related to location, property age, BHK configuration, furnishing, and construction status. These findings can support property screening, pricing review, and location-level monitoring. However, factors such as property condition, documentation, amenities, micro-location, negotiation conditions, and market timing may not be fully captured by the available data.

Overall, the project demonstrates how web-collected real-estate data, exploratory analysis, regression modeling, cross-validation, and statistical anomaly detection can be combined to create a practical decision-support framework for residential property-price screening. The model should be periodically updated with newer listing data as real-estate market conditions change.

7.2) References

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